



LCS Analyzer for Pairwise DNA Sequence

Name: Rokshana Akter Rubi
ID:2209004

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Name of the supervisor:Dr. Machbah Uddin
Department of Computer Science and Mathematics, BAU

ABSTRACT

This project introduces an LCS (Longest Common Subsequence) analyzer designed to compare biological sequences from different species and evaluate their pairwise similarities. Sequence data were sourced in FASTA format from the NCBI database. The application computes key metrics such as sequence lengths, LCS length and the similarity percentage between selected sequences. Additionally, a dot matrix visualization is included to provide a graphical representation of sequence alignments. A motif-finding tool is also incorporated to detect recurrent patterns based on user-defined motif lengths. The application offers insights into potential evolutionary relationships between organisms. By simplifying the complexity of sequence analysis, the tool facilitates genetic comparison and serves as an effective aid in bioinformatics education.

INTRODUCTION

This study explores the problem of DNA sequence comparison, highlighting its significance in genomics and evolutionary studies.

The primary objective is to develop an interactive application that visually and numerically compares two DNA sequences.

Key aspects include:

- LCS-Based Comparison – Identify the longest common subsequence between two sequences.
- Dot Matrix Visualization – Provide graphical matching evidence.
- Motif Detection – Find common k-length motifs between the sequences.

Through this research, we aim to contribute to accessible, educational tools for learning and applying bioinformatics concepts.

OBJECTIVES

- To analyze DNA sequence similarity using the Longest Common Subsequence (LCS) algorithm for identifying evolutionary relationships between organisms.
- To develop a user-friendly desktop application with a graphical user interface (GUI) using Python Tkinter that allows users to input and compare biological sequences interactively.
- To implement computational methods for calculating sequence length, LCS length and content, and similarity percentage between two sequences.
- To provide educational and practical value by making DNA comparison intuitive for students, researchers, and educators.

METHODOLOGY

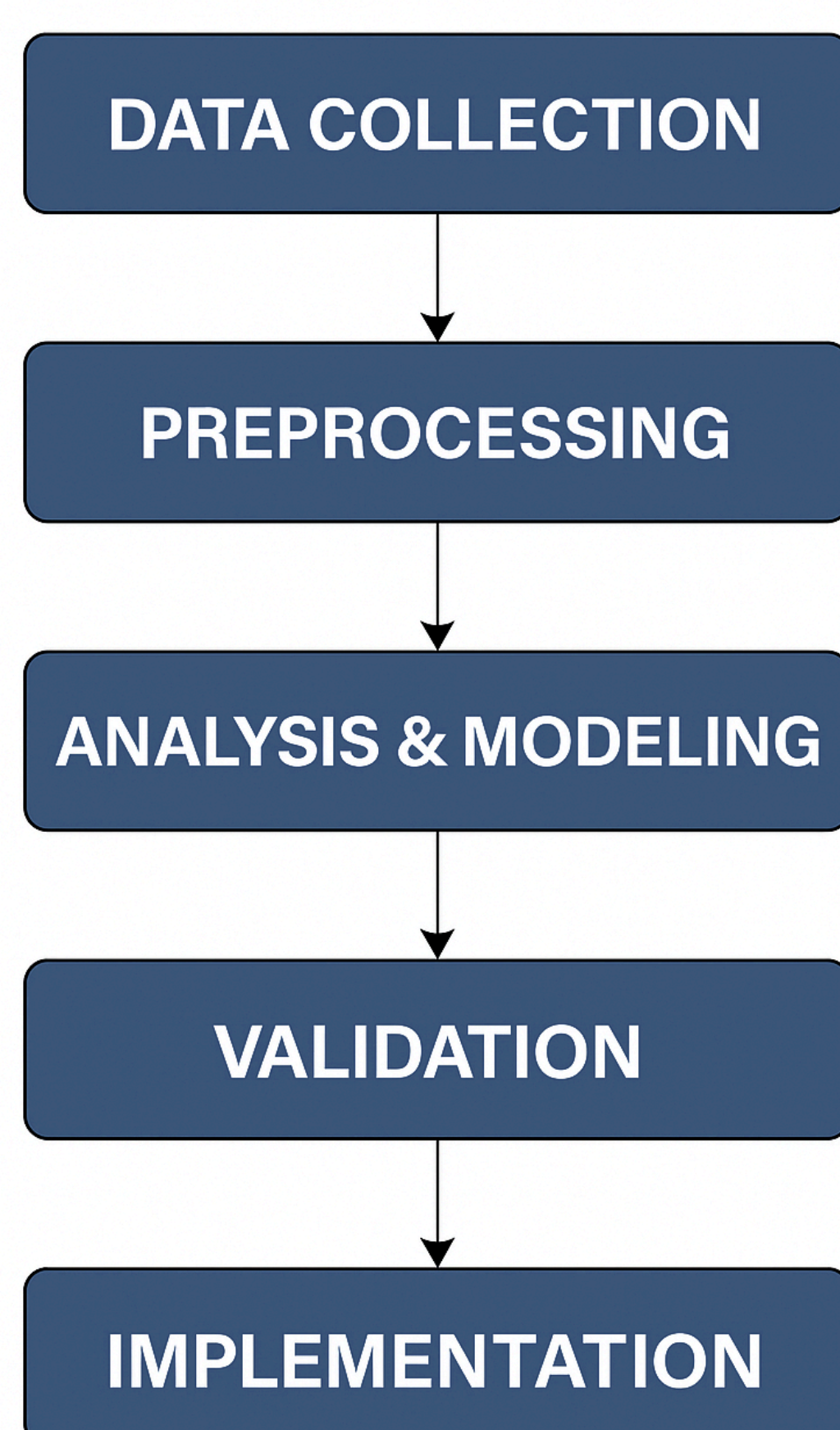


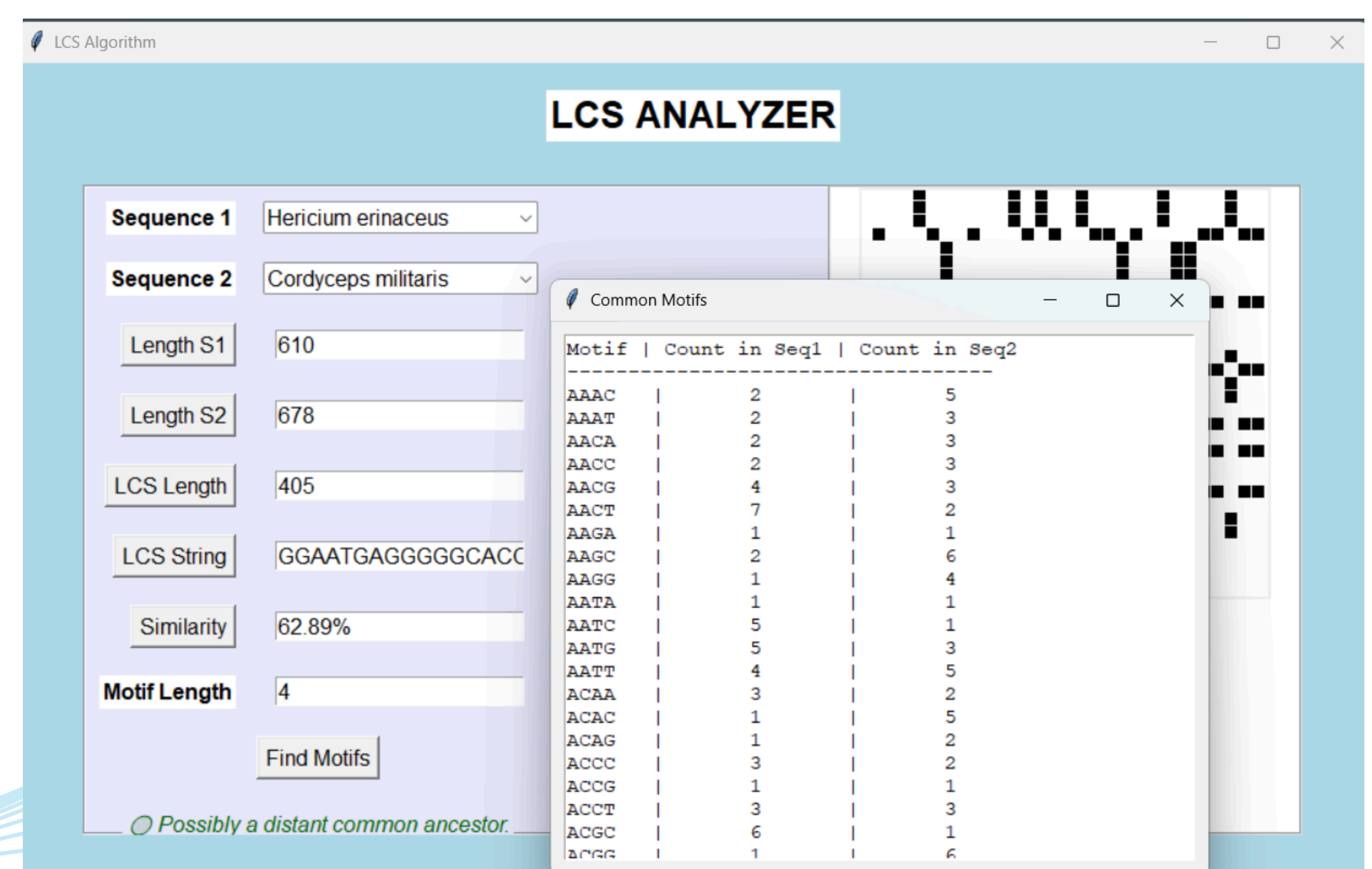
Fig. 1: Overview of the Method

This study follows a structured approach to develop a user-friendly DNA comparison tool using the LCS algorithm.

- Data Collection – DNA sequences were collected from NCBI in FASTA format and converted into key-value input format.
- Preprocessing – Sequences were cleaned and reformatted for consistent pairwise comparison.
- Analysis & Modeling – LCS algorithm implemented with Python and Tkinter GUI; results include LCS length, string, and similarity.
- Validation – Compared sequence similarity outputs with known results and manual inspection.
- Implementation – A fully functional desktop application was built for real-time comparison and motif finding.

This structured methodology ensures accuracy, interactivity, and clarity in DNA comparison results.

RESULTS



- The application successfully computes LCS length and similarity percentage between any two selected DNA sequences.
- Common motifs can be identified based on a user-defined length, with frequency counts displayed.
- The dot matrix visualization provides an immediate visual cue of sequence similarity.
- The similarity percentage offers basic insights into evolutionary relationships

KEY CONTRIBUTIONS

- Development of a user-friendly bioinformatics tool for pairwise DNA sequence comparison using the Longest Common Subsequence (LCS) algorithm.
- Integration of visual and analytical features, including dot matrix visualization and similarity percentage calculation, to enhance understanding of sequence alignment.
- Implementation of a motif-finding function that detects shared patterns between sequences based on customizable motif lengths.

CONCLUSION

The developed LCS Analyzer simplifies the process of DNA sequence comparison by integrating LCS computation, motif analysis, and visual representation into a single application. It provides a practical tool for exploring genetic similarity and can be used effectively in educational settings to support learning in bioinformatics.

REFERENCES

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